

Guidelines for encoding proteogenomics results

```
<SpectrumIdentificationProtocol analysisSoftware_ref="ID_software" id="SearchProtocol_1">
  <SearchType>
    <cvParam cvRef="PSI-MS" accession="MS:1001083" name="ms-ms search"/>
  </SearchType>
  <AdditionalSearchParams>
    <cvParam cvRef="PSI-MS" accession="MS:1001211" name="parent mass type mono"/>
    <cvParam cvRef="PSI-MS" accession="MS:1001256" name="fragment mass type mono"/>
    <cvParam cvRef="PSI-MS" accession="MS:1002490" name="peptide-level scoring"/>
    <cvParam cvRef="PSI-MS" accession="MS:1002496" name="group PSMs by sequence"/>
    <b><cvParam cvRef="PSI-MS" accession="MS:1002635" name="proteogenomics search"/></b>
  </AdditionalSearchParams>
</SpectrumIdentificationProtocol>
```

A

```
<PeptideEvidence dbSequence_ref="dbseq_generic|A_ENSP00000354925|" peptide_ref="DVLEGDSSSEDR_" start="23"
end="33" pre="A" post="A" isDecoy="false" id="DVLEGDSSSEDR_generic|A_ENSP00000354925|_23_33">
  <b><cvParam cvRef="PSI-MS" accession="MS:1002640" name="peptide end on chromosome" value="156646808"/></b>
  <b><cvParam cvRef="PSI-MS" accession="MS:1002641" name="peptide exon count" value="2"/></b>
  <b><cvParam cvRef="PSI-MS" accession="MS:1002642" name="peptide exon nucleotide sizes" value="25,8"/></b>
  <b><cvParam cvRef="PSI-MS" accession="MS:1002643" name="peptide start positions on chromosome"
value="156646122,156646800"/></b>
</PeptideEvidence>
```

B

C

```
<DBSequence searchDatabase_ref="SearchDB_1" accession="generic|A_ENSP00000389898|"
id="dbseq_generic|A_ENSP00000389898|">
  <cvParam cvRef="PSI-MS" accession="MS:1002637" name="chromosome name" value="1"/>
  <cvParam cvRef="PSI-MS" accession="MS:1002638" name="chromosome strand" value="+"/>
  <cvParam cvRef="PSI-MS" accession="MS:1002644" name="genome reference version" value="Ensembl release
84"/>
</DBSequence>
```

D

Feature	Explanation
A	If a proteogenomics search has been performed, this cvParam MUST be present.
B	Every PeptideEvidence that has isDecoy="false", MUST have either MS:1002740 "unmapped peptide" (for cases where a peptide could not be mapped) or the cv terms in bold MUST be present. For PeptideEvidence elements with isDecoy="true", all terms are OPTIONAL. In this example, peptide DVLEGDSSSEDR crosses an exon boundary. The N-terminal region of the peptide is mapped to positions 156646123 – 156646148 (start +25 from peptide exon nucleotide sizes). The C-terminal region of the peptide is mapped from 156646800 to 156646808 (second value of "peptide start positions on chromosome" +8). Definitions of terms are provided below
C	Additional CV terms MAY be added at a later date to encode classifications of peptide types, such as "novel junction", "novel N-terminus" and so on. Such information MAY be encoded on SpectrumIdentificationItem, using the peptide-level scores type of encoding.
D	Each DBSequence value MUST store either: 1) the genome reference version, chromosome name and strand or 2) be annotated with the term MS:1002741 "unmapped protein".